

Chapter – 8: A Journey Through States of Water

If it does not rain well, even the mighty ocean will be drained. – Thirukkural

- Bright summer day – Aavi, Thirav – enjoy *shikanji* (lemonade)
- Looking at ice – Thirav wonders – nature of ice, water
- He thinks – ice, water – diff. substances – Aavi thinks – they are same
- Aavi proposes – put some water in freezer – converts to ice
- Thirav explains – water is left in freezer – something added to it – converts to ice
- Is he right?
- Activity –
 - Put ice cube – in a cup – leave it on table – obs.
- Ice – converted to water
- Ice, water – same substance – diff. forms – called states
- Diff. states – show differences – their behavior
- Water – flows, splashes – ice – does not

Investigating Water's Disappearing Act

- Rainy morning – Aavi, Thirav – obs. – water puddles – playground
- Evening – while playing – surprised – some water – puddles – disappeared
- After washing – water on utensils – dries up
- Aavi thinks – water – seeps through utensils – just like soil
- Thirav thinks – it does not seep through utensils
- Activity –
 - Take – table spoon of water – steel plate
 - Obs. – whether – water seeps through – other side
 - Keep obs. – until water disappears
- Water – does not seep through – BUT – what happens
- Water – converts to gaseous state – **water vapours**
- Making dosa – sprinkle water – hot pan – it disappears
- Water – on hot pan – converts to steam – water vapour – some part – converts to water droplets
- Process – conversion – water to vapour – **evaporation**
- This process – continuous – even at room temp.
- Drying – wet clothes, mopped floor, body sweat

Another Mystery

- Next day – Aavi, Thirav and friends – decide – make lemonade
- Take – cold water – tumbler – add ice cubes
- After few minutes – notice something exciting – outer surface – tumbler
- Activity –
 - Take – cold water – glass tumbler – add ice cubes
 - Leave it – undisturbed – 5 minutes – obs.
 - **Record** obs. – ques. – touch outer surface – feel any change
- One obs. – Aavi's mind – some water droplets – outer surface – glass tumbler

- Initially – droplets deposit – combine together – form bigger drops
- Suggest – possible reasons – appearance of water droplets
- Aavi, Thirav – argue – chain of reasons –
 - Some water – seeped out – BUT – level of water – inside – same
 - Might have decreased – BUT – not significant enough – observable
 - Tall, narrow bottle – decrease in water level – observable
- Might see – dew drops – plants – mornings
- Boil water – half-filled utensil – cover it with steel plate – obs. – water droplets on plate
- Water vapour – inside air – contacts – cold surface – forms water droplets
- Process – conversion – water vapour to liquid state – **condensation**
- Activity –
 - Take – glass tumbler – half-filled water – few ice cubes – cover it – steel plate – weigh it – digital balance
 - Obs. reading – record weight – every 5 minutes
 - Keep obs. – 30 minutes – record it
- Predict** – mass of cold water – increase, decrease, remains same
- Water vapour – air – contacts – cold water – tumbler – converts to droplets – increase in readings
- Water – do not seeps through tumbler
- Water outside tumbler – due to condensation – cannot say exclusively – this activity
- Repeat above activity – BUT – mark the level – permanent marker
- This shows – water – not seeping – extra water – due to condensation

What are Different States of Water

- Water – substance – obs. – 3 diff. states
- Solid state – exists as ice – on heating ice melts – converts to liquid – further heating – converts to gaseous state
- Activity –
 - Take – ice cube – one container – transfer to another container – obs. change in shape – if any
 - Take – water – one container – transfer to another container – obs. change in shape – if any – also obs. – behavior compared to ice
 - Pour water – clean surface – obs. spread
 - Convert water to vapour – obs. spread – compare with water
- Ice (solid)** – retains its shape – whatever container – does not flow, spread
- Water (liquid)** – flows, changes its shape – not fixed shape – volume – constant – also spreads
- Vapour (gaseous)** – spreads – entire space available – no fixed shape – present in air – not visible at room temp.
- Other substance – also exhibit these states – wax, oil, ghee
- Ex. –
 - Solids – stones, wood, glass
 - Liquids – milk, oil
 - Gases – smell of cooked food, fumes from incense sticks, oxygen, air

How can We Change States of Water

- So far – water – exists – 3 states – solid, liquid, gaseous

- Change ice to water, water to vapour – provide heat – BUT – water to ice – what to do?
- Place water in freezer – converts to ice – take ice out – melts – converts to water
- Candle – made of wax – light it – melts – converts to liquid wax
- Cool this liquid wax – converts to solid wax
- Process – conversion – solid to liquid – **melting**
- Process – conversion – liquid to solid – **freezing**

How can Water be Evaporated Faster or Slower

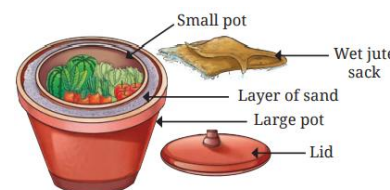
- Obs. surroundings – conditions – affecting – speed of evaporation
- Activity –
 - Take – water – small cap – bottle
 - Same amt. – plate – exposed area – diff.
 - Keep them – near each other
 - Record – time taken – evaporate – each case
- Water in plate – faster evaporation – larger exposed area

Other conditions which affect how fast water evaporates

- Activity –
 - Take – 2 identical bottle caps
 - Pour water – same amt. – each cap
 - Place one cap – sunlight – other cap – shade
 - Obs. – every 15 minutes
 - Record obs. – time taken – evaporate completely
 - Repeat activity – windy, rainy day
- Conclusions –
 - Faster evaporation – under sunlight, windy day
 - Slower evaporation – rainy day, higher **humidity**

Cooling Effect

- Aavi's mother – replaces stainless steel pot with *matka* (earthen pot)
- Returning from school – Aavi – drinks water – surprised – so cool
- She thinks – why is water so cool – why doesn't it cool inside stainless steel pot
- Water – seeps – surface of earthen pot – evaporates – imparts cooling effect – water inside it
- Activity –
 - Take – 2 earthen pots – diff. sizes
 - Fill – sand – larger pot – bottom
 - Place the smaller pot – inside larger pot
 - Fill the gap – more sand – add water
 - Cover – top – smaller pot – wet jute bag
- Allow 4-5 hrs. – pot-in-pot cooler – cools down
- Obs., discuss – cooling effect created by it
- Place fruits, veggies – smaller pot – obs. – 1 week – freshness
- Add water – regularly – keep the sand – moist
- Some houses – use *surahi* – unique clay pot – keeps water cool



How Do Clouds Give Us Rain

- Condensation – imp. role – bring back – evaporated water
- Air – moves higher – earth's surface – gets cooler
- Certain height – air – so cold – vapours inside – convert to droplets – wrap around dust particles
- Small droplets – float in air – form clouds
- Many droplets – combine together – bigger drops – some drops – so heavy – start falling – called rain
- Special condition – falls as hail, snow
- Activity –
 - Take – empty 1 litre plastic bottle – pour 1 cup water
 - Close lid tightly – squeeze, release 2-3 times – quickly – obs. empty space – inside bottle
 - Repeat activity – place burnt piece – newspaper – inside bottle – obs.
 - Burnt paper – provide dust particles – vapour attach to them – form clouds
- Water from ocean, earth's surface – evaporates – vapours – returns – rain, hail, snow – flow back – oceans
- This circulation – **water cycle**
- Small portion – available water – fit for use – plants, animals, humans
- Most water – oceans – cannot use directly –
- Population increases – demand increases – creates shortage
- Very imp. – use water carefully – without wasting